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ORIGINAL ARTICLE

Energy assessment of the Portuguese meat industry

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Abstract The meat product industries play a major role in many global economies. In several countries, it is the industry with a higher economic weight within the food industries. As refrigeration systems are indispensable within the production processes, it is important to improve the overall energy efficiency in order to reduce the electricity consumption in these industries. The electrical energy consumption of the Portuguese meat industries is evaluated in this paper. The field study shows that electrical energy accounts for 66.5 % of the overall energy consumption in slaughterhouses, 85 % in sausage houses and 92.5 % in ham industries. Relatively to the refrigeration systems of these industries, results show that the average compressor nominal power per unit of cold room volume is comprised between 0.072 and 0.043 kW/m^3. The average value of the specific electrical energy consumption for the slaughterhouses was 149 kWh/ton$_{HSCW}$, 660 kWh/ton$_{RM}$ for sausage houses and 1208 kWh/ton$_{RM}$ for ham production industries. Finally, a potential reduction of the electricity consumption based on simple energy efficiency measures was estimated in 24 % for the slaughterhouses, 13 % for the sausage houses and 8 % for the ham industry.

Keywords Energy consumption · Energy efficiency · Electrical energy · Refrigeration · Cold chambers · Meat industry

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Introduction

Consumer preference regarding the use of meat in their diet has grown in recent years. Between 1980 and 2005, the annual world average per capita consumption increased by 37.3 %, with a peak in 2005 of 41.2 kg/capita/year (FAO 2009). Consequently, the global production of meat has also increased and is expected to reach 470 million tons in 2050, more than the double of the 229 million tons verified in 1999/2001 (Nigel et al. 2010). Currently, the meat product industries play a major role in many global economies. In several countries, it is the industry with a higher economic weight within the food industries. The manufacturing industries within which stands the food industries belong to the industrial sector that are globally responsible for 37 % of

a worldwide energy consumption and 36 % of carbon dioxide emissions (CO_2) (Abdelaziz et al. 2011). At the European level, the industrial sector accounts for 24.9 % of the total energy consumption with the food industry representing 10.1 % of the energy consumption of the EU industrial sector (Eurostat 2015). In fact, the EU food industry is the largest manufacturing sector (14.6 %) and represents a universe of 286,000 industries with 4.2 million direct jobs and had in 2014 a turnover of 1048 billion euros (FoodDrinkEurope 2014). In Portugal, the food industries follow the European trend with the industrial sector accounting for 29.6 % of the total energy consumption and with the food industry representing 8.9 % of the energy consumption of the Portuguese industrial sector (Eurostat 2015). Moreover, the food industries are also the largest group of manufacturing industries in the country (13.2 %). Within the food industry, the largest sector corresponds to the meat industry. In 2009, this sector was composed of a set of 619 licensed industries (INE 2011) and having achieved a sales volume of 21.3 % considering the 8267 million euros obtained by food industry sectors (INE 2010). The growth of this sector has been promoted by the increase of sausage production. In 2009, these products represent 24.5 % of the 279265.9 tons of meat products production, with a strong contribution from the ham industry, which in this year increased 9.3 %, compared to the previous year (INE 2010). This evolution follows the trend of many European countries (Wilcock et al. 2004; Röhr et al. 2006). with particular emphasis on the Mediterranean ones due to their long tradition in the manufacture of meat products (Toldrá 2003; Ray 2003; Gurbuz et al. 2009; Jiménez-Colmenero et al. 2009). The production of sausages and ham in Portugal is mainly industrial, although based on traditional elaboration processes. Changes introduced in the process of traditional elaboration relate mainly to the mechanisation of processes and the use of drying and/or smoking mechanisms with temperature and humidity control. Thus, it enables a continuous production throughout the year, without the influence of climatic conditions, allowing its maturation to take place in any geographic area.

Refrigeration systems are indispensable within the production processes of the meat industry in order to prevent changes in meat composition, bacteria action and development of microorganisms, and to obtain stabilisation plus the organoleptic characteristics of the products (Savell and Mueller 2005; Toldrá 2002). In slaughterhouses, refrigeration systems are also used to carry out the cooling of bovine, swine, sheep and goat carcasses, of an initial value ranging from 38 to 40 °C to a value close to 7 °C, or lower, for meat, or 3 °C for viscera, within a period of 24 h after slaughter, and subsequently for its conservation chilled or frozen, at a temperature between 0 to 2 °C and −18 °C, respectively (Bowater 1997; James and James 2002). Sausage houses and ham manufacturing industries use powerful refrigeration units and cold rooms with computer-controlled temperature, air velocity and relative humidity for the meat preservation process, ripening and drying of products with values of temperature and relative humidity comprised between 0 to 30 °C and 65 to 85 %, respectively (Vázquez and Vanaclocha 2004).

The use of refrigeration systems is primarily responsible for the electricity consumption in these industries and may represent 60 to 90 % of electricity use in slaughterhouses (Marlow and Colley 2007; Pagan et al. 2002; European Commission 2003) and from 40 to 50 % for other categories of meat processing industries (Ramírez et al. 2006; Fritzson and Berntsson 2006; Alcázar-Ortega et al. 2012). Furthermore, it should be noted that electricity costs might correspond to two thirds of the total energy costs in meat processing industries (HTC 2009). In Portugal, there is a similar trend in the food sector, with an increase of 14 % of electricity consumption, between 2008 and 2010, having achieved 1599 GWh in 2010, corresponding to 8.8 % of the electrical energy consumed in the industrial sector and 3.2 % of electrical energy consumed in Portugal that year (DGEG 2011). This increase in energy consumption occurs in a context of general rise of the fuel prices (in Portugal, between 2007 and 2011, the price of electricity and natural gas increased by 24 and 14.4 %, respectively), having a significant impact on the competitiveness and sustainability of Portuguese meat industries. The current concerns about the scarcity of the energy resources together with environmental impacts resulting from their use require specific measures to improve energy efficiency and to minimise the usage of fossil fuels (Almeida and Silva 2011; Nunes et al. 2011, 2014). In order to mitigate these effects, government diplomas with energy-saving measures are being implemented to improve energy efficiency in various sectors of economic activity, with special emphasis on the Portuguese industrial sector. However, these government diplomas are directed towards facilities of intensive energy consumption, being those that present an

annual consumption of primary energy exceeding 500 tons of oil equivalent (toe), leaving out more than 90 % of food industries, with those being represented by micro and small enterprises.

The present study characterises the energy consumption profile of meat industries that consume less than 500 tons. It particularly assesses the consumption of electric energy (energy which is mostly used by these industries), evaluates specific electrical consumption indicators, presents the potential of energy saving and a set of appropriate measures to improve their energy efficiency. This study intends to contribute positively towards a better understanding of the meat energy consumption profile in Portugal.

Productive processes

Meat production in slaughterhouses

Figure 1 presents the typical process of cow, pig, sheep and goat slaughter in Portugal, which essentially comprises the following operations: reception, ante-mortem inspection, animal slaughter, carcasses and offal cleaning, post-mortem inspection, cooling, cutting, processing and dispatch. For each animal species, there is a specific circuit where carcasses are moved through air rails into fast cooling tunnels and refrigeration or freezing chambers.

Rapid cooling is the most used method in Portuguese industries and takes place in two complementary steps. The first consists of rapidly lowering the initial temperature of 38 °C/40 °C to a temperature around 15 to 20 °C. The second step, relates to final stabilisation of the meat's temperature to a value below 7 °C in the centre of the meat. The first step takes place in a rapid cooling tunnel or in a cold chamber with strong air circulation. In both cases, the temperature and relative air humidity inside lies between −1 to 2 °C and 82 to 90 %, during approximately 4 h for cow carcasses and 2 h for pig and sheep. The second step takes place inside cold rooms with lower air circulation and with environmental conditions similar to those above. These two steps are similar to those described in James and James (2002). In Portugal, the slow cooling of pig carcasses is a rarely used method due to appreciable weight losses and possible dangers of microorganism growth (Bowater 2001). In this method, carcasses are left exposed to the room temperature within the premises, until they reach a temperature of 30 °C, and then are taken to the interior of cold rooms with a temperature ranging from 0 to 4 °C, relative humidity close to 80 % and weak air circulation being necessary around 24 h to obtain the temperature stabilisation of the carcasses. This method has the advantage of involving lower power consumption. Freezing meat is a procedure carried out, most of the time, with prior refrigeration (biphasic freezing) and is held in freezing chambers or tunnels with relative humidity and temperature values presented in Fig. 1. Although freezing is not frequent in Portuguese slaughterhouses, when it occurs, it requires the control of the relative humidity inside the cold room in order to avoid weight loss of the product (Bowater 1997). The one-step freezing (monophasic freezing), is also used in some industries, through quick-freezing tunnels. The final product is kept in cold rooms (temperature between −1 and 2 °C) or freezing (temperature of −18 °C) and may thereafter go to the cutting room (only the refrigerated product) or be dispatched.

Traditional sausage production in sausage houses

In Portugal, there is a great diversity of traditional sausages, produced with different raw materials and ingredients, but they all have the production process in common. This process is based on three fundamental steps (Toldrá 2002). the first is the stabilisation of raw materials, achieved in the so-called cold phase (choosing, cutting, spicing and dough preparation); the second is the development of sensorial characteristics that occur during ageing or maturation phases; and finally, the third consists of thermal treatment of products to cause dehydration and intensification of texture, aroma and flavour of the sausage.

Figure 2 shows the layout of the manufacturing process of sausages in Portugal, which comprises the following steps: receipt of raw material, preserving the meat in a refrigerator/freezer, choosing, cutting, spicing and dough preparation, maturing, filling and drying, stabilisation and final packaging and shipping. Cutting operations, mixing and filling are held in air-conditioned rooms with a temperature of 12 °C maximum. In the maturation phase, the mixture of meats with its condiments is kept in cold rooms, with temperatures between 1 and 6 °C and a relative humidity of 80 to 85 %, from 24 to 48 h. The drying process of sausages is accomplished by controlled atmosphere drying or drying through smoke. Controlled air-drying consists of

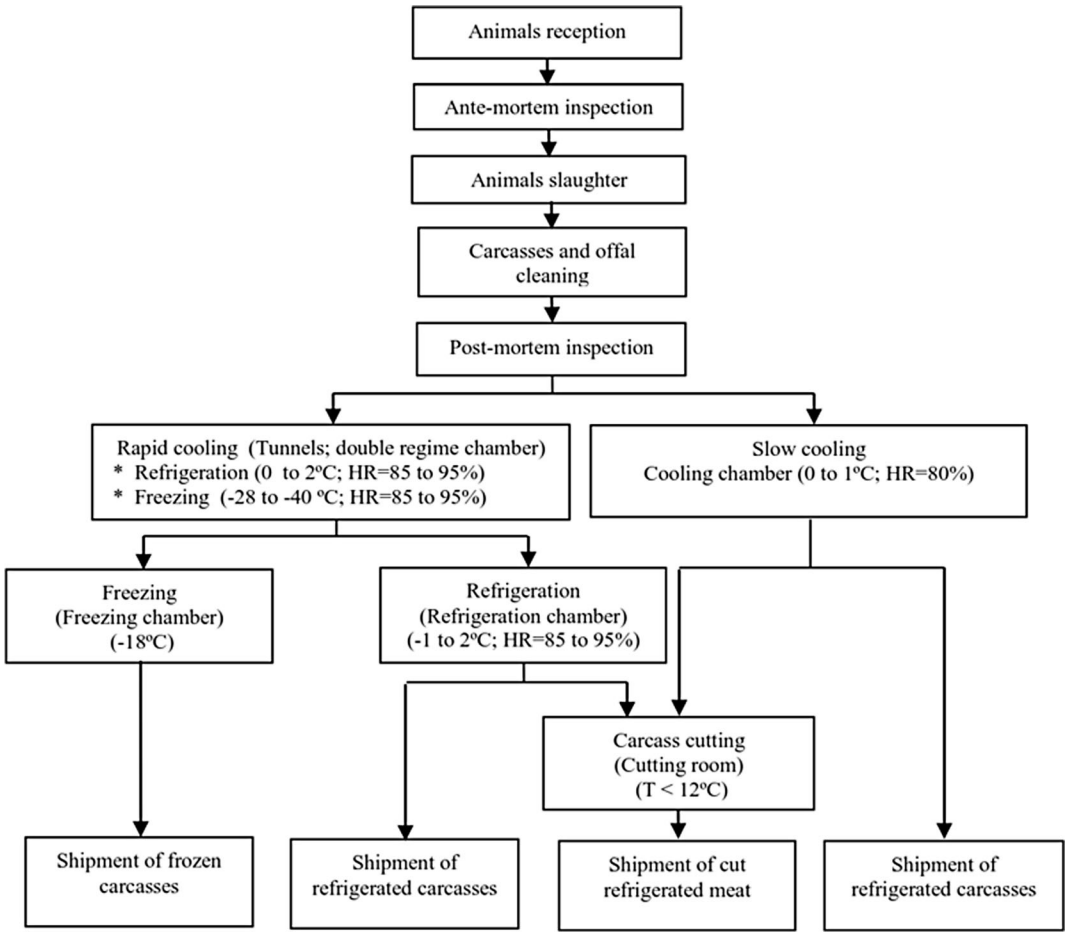


Fig. 1 Manufacturing diagram of animal slaughter

placing sausages in rooms with temperature and humidity control. The duration varies according to the dimensions of the products, typically between 5 and 15 days for thin products and 30 to 60 days for thicker products (Arnau et al. 2007). In Portugal, the final product presents a level of relative humidity from 30 to 40 %. When drying with smoke, products are exposed, during 3 to 5 days, to the action of heat and smoke resulting from combustion of wood (oak, in Portugal). In both cases, after drying, sausages are placed in a cold chamber for final stabilisation.

Ham manufacturing in the meat processing industries

Ham is a product obtained artificially in cold rooms from the ageing of the hind legs of white pigs during an approximate period of 6 months. The production process consists of a set of operations with different values of temperature, relative humidity and air circulation. After the first stages of leg preparation, which involves thawing, weighing, pressing and kneading, held in environments with controlled temperature and relative humidity, the main stages of the production process, include salting, post-salting, drying and stuffing (Toldrá 2003). Salting is intended to provide the proper amount of salt to the legs, keeping them involved in salt from 8 to 12 days. This phase takes place inside the cold rooms with high humidity and low temperature (Arnau et al. 2007). In the post-salting phase, the salt diffusion towards the inner part of the leg takes place from 30 to 45 days. At this stage, the legs remain inside the cold rooms to ensure low air temperature and high humidity (6 °C and 85 % relative humidity). In the next phase, the dehydration of the leg takes place slowly during a period of approximately 45 days. The temperature in the cold chamber is gradually increased until it reaches a value close to 14 °C, and at the same time its relative humidity is lowered slowly until it reaches

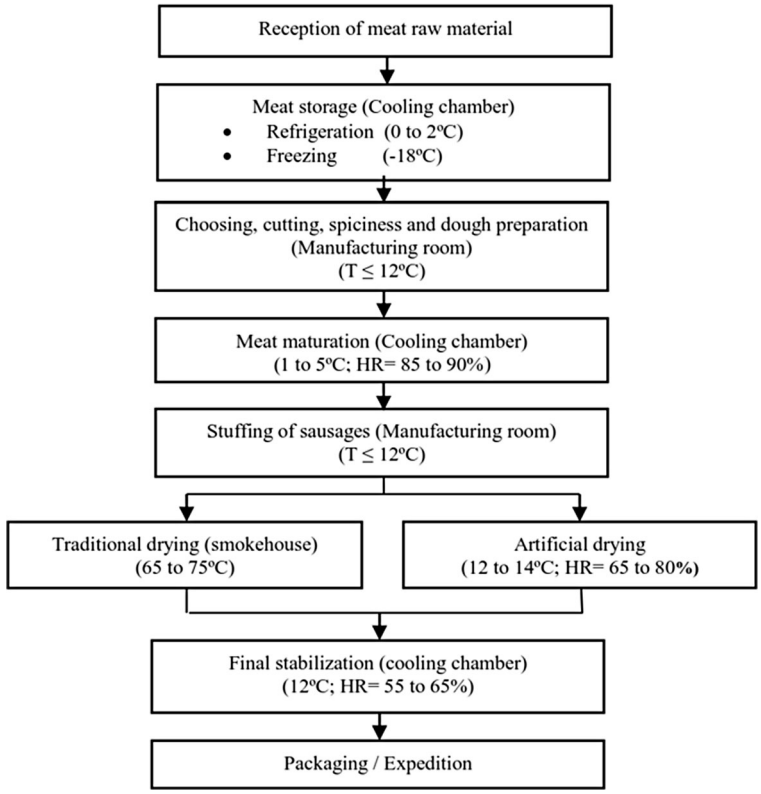


Fig. 2 Scheme of the sausage production process

approximately 75 %. Stuffing finishes the drying process and assigns special organoleptic characteristics of the ham for a minimum period of 15 days. In this phase, the temperature inside the room gradually increases up to a maximum value (about 26 °C in Portuguese industries) and the relative humidity is kept close to 75 %. After finishing the manufacturing process, hams are kept inside cold rooms for final stabilisation during a minimum period of 15 days. The production process is carried out inside cold rooms, some with refrigeration systems for maintenance of products at low temperatures and others with specific refrigeration systems (compact systems, with cooling and heating batteries) for controlling temperature and relative humidity inside the rooms (drying and stuffing process). Figure 3 shows the scheme of the typical ham manufacturing process applied to Portuguese industries.

Material and methods

In order to evaluate the electricity consumption profile of meat processing industries, a survey was carried out to 33 selected manufacturing units located in the interior region of Central Portugal. The selected companies (in particular, ham and sausage industries) follow traditional meat manufacturing processes and are well representative of this kind of industry in Portugal. The details of the selection process can be found in Nunes (2014). According to the main activities and products developed, they were classified into three categories: slaughterhouses (mainly pigs, 4 units), sausage houses (20 units) and the manufacturing industries of ham (9 units). The survey information had two phases: filling out a survey and measurement of parameters. The aim of the survey was to compare the infrastructure characteristics (location, age, type of materials, dimensions and location of cold rooms), activities (type and quantity of raw materials and products, production processes), characteristics of refrigerating systems (age, location, type of technology, cooling system and electrical power), cold rooms conditions (temperature and relative humidity) and energy consumption (electricity, natural gas and other fuels – based on monthly invoices). The surveys were used to gather information regarding the production process as well as quantifying

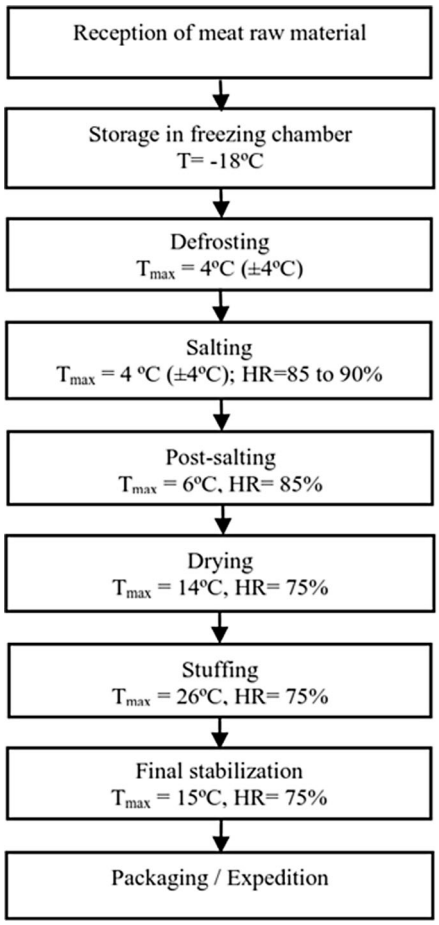


Fig. 3 Scheme of the ham production process

temperatures, relative humidity and time expended in the various stages of the process.

In the second phase, visits to industries were carried out with the aim of measuring some relevant parameters. Environmental conditions of the interior and exterior of cold rooms were measured, namely the temperature and relative humidity through a multifunction digital equipment (Testo 435–2), and probe with a temperature range of −20 to 70 °C and 0 to 100 % for relative humidity, with an accuracy of ±0.3 and ±2 % RH. To evaluate thermal losses, the temperature of the outer surface of cold rooms was measured; first, through a general observation with infrared thermography equipment (Testo 880–4; ±2 °C accuracy) and later, in the most sensitive points, with the digital equipment (Testo 435–2) and a contact probe with thermocouple (type k), with a temperature range of −60 °C to 300 °C and an accuracy of ±2 %. The interior dimensions of the cold rooms were determined with a range finder using the infrared technique (Bosch-DLE 40), with a range of 40 m and an accuracy of ±1.5 mm. In particular, for the mechanical refrigeration system, temperatures were measured at the compressor inlet and evaporator outlet, to assess the degree of superheat. The overall electricity consumption of the plants was also evaluated as well as the electricity consumption of the compressors. This measurement was carried out through a digital energy analyser (Elcontrol-Energy Explorer), for a range of values of 15 to 750 V and 20 to 1000 A, with an error margin of 0.53 to 2 V and of 0.04 to 2 A. Electrical power absorbed by the compressor was determined by measuring the voltage values and intensity of electrical current, measured with a digital ammeter clamp (Escort ECT-620). The value obtained was compared to information from catalogues and manufacturers' calculation software, based on the measured evaporation and condensation temperatures and superheat and sub-cool degree. Close results (deviations below 10 %) were found between the two methods in use.

In order to compare the performance of different industries presented in this study, the following three indicators were introduced: specific energy consumption (SEC), volume usage rate of the cold rooms (EVU) and rate of electrical power absorbed by compressors per unit volume of cold rooms (EPCV). The specific energy consumption (SEC) is a physical indicator usually referred in literature to evaluate the energy efficiency of industrial units or sectors (Maxime et al. 2006; Nanduri et al. 2002; UNIDO 2010). This indicator was recently used in energy analysis of meat industries, with a particular focus on the industries of animal slaughter and for energy evaluation of meat industries in different European countries (Ramírez et al. 2006). The specific energy consumption (SEC) was evaluated by Eq. (1).

$$SEC = \frac{E}{RM} \tag{1}$$

Where E represents the annual consumption of electrical energy, RM designates the annual production of hot carcasses (ton_{HSCW}) in the slaughterhouse or the annual quantity of raw material processed (ton_{RM}) for manufacturing industries of sausages and hams.

Another indicator calculated in this research was the effective volume usage of the cold rooms (EVU). This indicator is given by Eq. (2).

$$EVU = \frac{RM}{V_{\text{total}}} \tag{2}$$

Where, V_{total} indicates the total volume of the industry's cold rooms. With this indicator, it is possible to evaluate the handling level of raw material in cold rooms. Thus, within the same category of industries (using the same productive process), it estimates the greater or lesser use of cold rooms with the consequent implications at the level of energy consumption. Finally, another calculated indicator was the electrical power absorbed by the refrigeration compressors per unit volume of the cold rooms (EPCV) which is given by Eq. (3).

$$EPCV = \frac{P_{\text{total}}}{V_{\text{total}}} \tag{3}$$

Where P_{total} indicates the overall compressor power consumption of the industry. This indicator gives us an idea of the cooling capacity installed per unit volume: when used in industries that produce the same product (i.e. in the same category) by the same process, it can indicate quality of cooling system design; when applied to industries of the same sector but with different products (different categories), it indicates the cooling requirements for each meat product (a relevant piece of information for industries that intend to change their products or decide to add another product using the same cold rooms).

Results and discussion

This section sets out the results of general information collected in 33 meat industries (slaughterhouses, sausages and hams) and encompasses the information of 355 cold rooms, 2 fast cooling tunnels and 226 refrigeration systems. Later, there will be a separate analysis of results regarding the production process. In general, the premises of Portuguese meat industries are comprised of buildings with one floor, brick walls and roof assembly (galvanised, fibre-cement or red clay tile). A common feature to almost all industries is the reduced physical space between the concrete slab, or suspended ceilings, and the roof (usually non-ventilated attic) therefore originating the occurrence of high internal temperatures (in summer days, we measured air temperatures up to 55 °C). Typically, the buildings are divided into several support rooms inside, some conditioned rooms for the manufacture process, several cold rooms and some freezing rooms. Almost all the walls and ceilings of cold rooms and conditioned spaces are made of insulated (sandwich) panels, with insulating core made of rigid polyurethane foam and coating on both sides with metal sheet. The thickness of the walls is quite variable, being 40 mm for conditioned rooms, 60 or 80 mm for cooling rooms and 80 to 100 mm for freezing rooms. The ground of the cold rooms consists of black agglomerate cork layers or high-density polyurethane boards (40 kg/m^3), including a vapour barrier (polyethylene film). In older industries, the walls of the cold rooms were built in brick masonry with cork agglomerate flooring. This kind of constructive solution presents insufficient thermal resistance (visible in thermal images) which penalises the energy performance.

The location of cold rooms within the premises is based on the type of production circuit process: reception, production, processing, ageing, drying, storage and dispatch. These industries have 3 cold levels: freezing (−18 °C), refrigeration (0 to 6 °C) and cooling (12 °C). The mechanical cooling systems are the vapour-compression refrigeration type. The difference between the values measured for the power absorbed by the compressor and the corresponding value indicated by the software/catalogue of the manufacturers are less than 10 %. In general, the evaporators are inside the cold rooms and have defrosting systems (where electrical devices are the most common). Condensation groups and compressors are usually found outside the premises (some of the condensation groups are exposed to direct sunlight, in close proximity, see Fig. 4) though there have also been situations where these elements are placed above cooling rooms in non-ventilated attics. The lack of equipment maintenance is illustrated in Fig. 5. In this situation, the supplying of cold air to the storage room is made from the outside of the plant. Thermal imaging allows us to highlight the gains resulting from sections of ducts with thermal insulation already damaged. It is important to refer that the location of these ducts are directly above the condensation groups. Any of these situations penalises the efficiency of the refrigeration system. In the following subsections the remaining characteristics



Fig. 4 Condensation groups in close proximity

of the infrastructures will be presented and discussed separately for each type of industry: slaughterhouses; sausage and ham houses. Later, a comparative analysis will be carried out.

Slaughterhouses

The visited slaughterhouses (4 units) were units with large facilities, generally horizontal type in which all slaughter operations are developed on the same floor. These slaughterhouses include reception rooms attached to the main building and specific air lines for each animal species to perform transportation along the production process. Table 1 indicates the number of cold rooms in each manufacturing plant (this includes fast-cooling tunnels, freezing rooms and cold rooms for chilled and frozen products). Two of the visited slaughterhouses (plants 3 and 4) have higher production values. The main activity of these plants is the production of meat, as whole or cut carcasses, chilled or frozen. Its main characteristic is the high movement of the products. Hence, the energy consumption associated to each product is therefore reduced. The main types of energy consumed by these plants are electricity for powering electric motors of the refrigeration systems, air lines for animal transport, electric saws, air compressors, lighting and fuels (natural gas, propane, heating oil) for water heating. The energy balance made to these industries indicates that, on average, 66.5 % of the energy consumed is electricity.

Table 1 summarises the information collected at slaughterhouses in the region and the results of the above-mentioned indicators. This table shows that the annual electrical consumption of slaughterhouses is from 208.4 to 1176 MWh, and that the overall compressor power lies between 28 and 260 kW. Specific consumption values of electricity for this category are presented in Fig. 6. Results indicate a SEC variation between different industries, from 84.3 to 249 kWh/ton_{HSCW}, with an average value of 149 kWh/ton_{HSCW}. The largest SEC difference is between plant 1 and plant 3. Plant 3 shows a value of this indicator superior to 57.4 % when compared to plant 1. In particular, plant 3 has older equipment and facilities, bigger dimensions, and cold rooms built in brick masonry coated with cork boards (instead of sandwich panels of polyurethane, as others plants). Although plant 3 has the greatest number of cold rooms, it was observed that they operate on partial load, therefore, presenting the lowest value of EVU. This poses an energetic penalty because it would be possible to make a more effective use of fewer cold rooms to treat the same production. Another unfavourable indicator for plant 3 is the EPCV. It presents the largest value of slaughterhouses studied and shows that there is an over-sized cooling capacity installed. Analysing the SEC behaviour we verify that there is a potential energy consumption reduction by

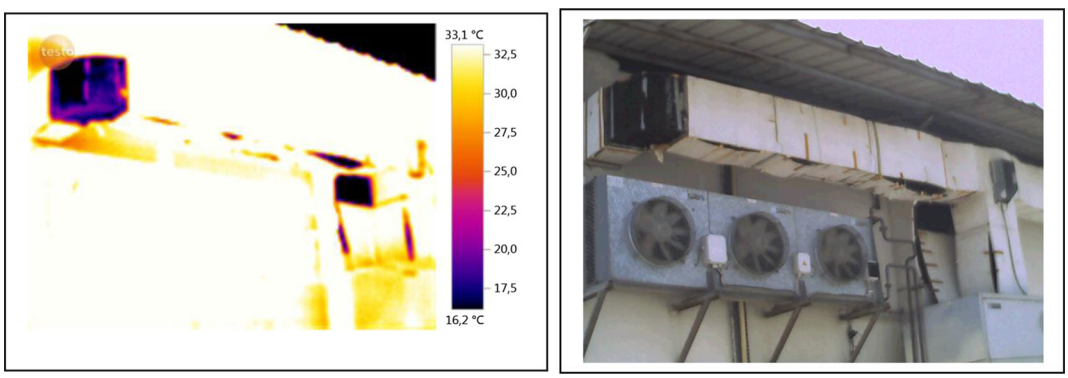


Fig. 5 Thermal image of the external ducts

Table 1 Results for slaughterhouses, based on the information collected locally in each plant and with the indicators calculated by Eqs. (1) to (3)

Plants	Year of construction	Annual production (ton_{HSCW})	N.^{er} cold chambers	V_{total} (m^3)	P_{total} (kW)	E (MWh)	SEC (kWh/ton_{HSCW})	EVU (ton_{HSCW}/m^3)	EPCV (kW/m^3)
Plant 1	1990	1473.7 (pigs+cattle)	9	638	43	209.2	142	2.31	0.067
Plant 2	2000	1751.1 (pigs)	9	517	28	208.4	119	3.39	0.054
Plant 3	1982	4723.1 (pigs+cattle)	24	2499	260	1176.0	249	1.89	0.104
Plant 4	1995	4402.8 (pigs)	18	1646	106	370.8	84.3	2.67	0.064

simply making industries that show specific consumptions above average to apply the good practices adopted by the others industries that, in the same category, featured the best energy performance. In fact, the potential savings in an industry corresponds to the difference between the value of the individual consumption of this industry and the average of the energy consumption in this category.

Sausage houses

Sausage houses are small industries with old facilities, five cold rooms in average (conservation chambers), one drying room and one smoke room. These 20 units visited are mostly family-oriented and located in regions with strong tradition of meat product manufacture and do not have a strong production. The room dedicated exclusively to drying, uses compact refrigeration units that allow the control of air temperature and humidity within the room, using secondary condensers (for air heating) and evaporators (for air cooling/drying). Energy consumption in these industries is electrical (operation of the components of the mechanical refrigeration systems, compressed air production, lighting and fans and equipment for cutting, mixing and filling) and thermal (water heating). The electrical component is the most relevant, representing, on average, 85 % of the energy consumption.

Table 2 shows the information concerning sausage houses. This table shows that the annual consumption of electrical energy of sausage houses is between 12.6 and 673.3 MWh and that the overall compressor power lies between 3.8 and 126.7 kW. This range of values is due to the different sizes of the industries in this category.

The SEC shows values between 248 and 1840 kWh/ton_{RM}. Plant 8, for example, with a specific energy consumption of 1840 kWh/ton_{RM}, the highest value registered, is above the average value of these industries, 660 kWh/ton_{RM} in this case, as depicted in Fig. 7. Although this company is not the oldest, it presents some special characteristics. It has the merit of scrupulously following the temperature process; however, it presents an inappropriate location of the condensation groups.

In fact, these groups are, oddly enough, located within the premises, in air-conditioned areas. In this particular case, it turns out that there is high energy saving potential. Another important aspect is the thermal resistance of the infrastructures (buildings envelope and cold rooms) in the energy performance of the industries, and it is illustrated by plant 10, one of which features smaller SEC. The buildings' walls are made of 120 mm polyurethane sandwich panels, as well as the refrigeration and freezing rooms, which have thicknesses of 100 and 120 mm. The roof also uses the same type of panel, with a thickness of 40 mm, and having a ventilated attic.

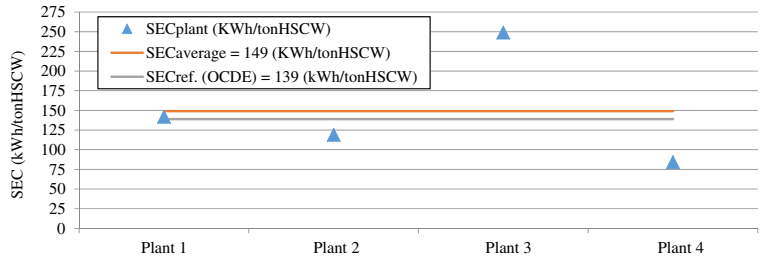


Fig. 6 Values of the specific electricity consumption (SEC) on each Portuguese slaughterhouse plant visited, the average value and a reference value from OECD countries (UNIDO 2010)

Table 2 Results for sausage plants, based on the information collected locally in each plant and with the indicators calculated by Eqs. (1) to (3)

Plants	Year of construction	Raw material (ton_{RM})	N.^{er} cold chambers	V_{total} (m^3)	P_{total} (kW)	E (MWh)	SEC (kWh/ton_{HSCW})	EVU (ton_{HSCW}/m^3)	EPCV (kW/m^3)
Plant 1	1998	38.5	4	80.0	7.8	23.7	616	0.48	0.098
Plant 2	1992	45.3	4	188.0	11.8	38.2	844	0.24	0.063
Plant 3	1997	104.8	5	172.7	8.6	44.1	421	0.61	0.050
Plant 4	1992	877.2	12	675.0	44.0	245.0	279	1.30	0.065
Plant 5	1990	76.5	6	248.6	19.0	68.7	897	0.31	0.076
Plant 6	2003	450.4	9	679.0	71.6	232.4	516	0.66	0.105
Plant 7	1998	149.9	4	126.0	17.8	47.4	316	1.19	0.141
Plant 8	1994	78.5	8	516.0	27.8	144.5	1840	0.15	0.054
Plant 9	1995	198.0	6	547.0	31.6	66.1	334	0.36	0.058
Plant 10	1995	315.8	6	321.0	27.2	82.8	262	0.98	0.085
Plant 11	1998	55.5	3	26.0	9.3	27.3	491	2.14	0.358
Plant 12	1995	1040.9	9	850.0	94.6	258.0	248	1.22	0.111
Plant 13	1982	178.3	7	866.0	55.4	194.1	1088	0.21	0.064
Plant 14	1990	31.1	2	14.0	3.8	12.6	405	2.22	0.271
Plant 15	1970	178.1	9	764.0	24.5	135.8	818	0.23	0.032
Plant 16	1993	80.3	4	173.0	13.3	39.5	491	0.46	0.077
Plant 17	2003	45.5	4	175.0	12.9	25.5	560	0.26	0.074
Plant 18	1995	68.0	4	122.0	4.2	33.5	494	0.56	0.034
Plant 19	1995	1003.0	23	2077.0	126.7	673.3	671	0.48	0.061
Plant 20	1986	92.3	9	510.0	21.4	148.7	1611	0.18	0.042

Extending the analysis to all industries and whereas the same procedure was adopted for the slaughterhouse industries, the potential for reducing the overall consumption of electricity would be around 13 % (322 MWh). Once again, it should be noted that this procedure assumes sharing the best practices in the sector.

If we take as a reference the average value of specific electrical energy consumption for the meat industries in Spain, in this case from 465 kWh/ton_{RM} (EREN 2008). the potential for reducing the electrical energy consumption is 27 % (683 MWh).

The reference value of the meat industry is also presented in Fig. 7. It should be noted that spanish industries incorporate in the productive process other forms of energy in addition to the electrical one (EREN 2008). Moreover, the drying process of Portuguese sausages rarely takes place in natural environments. These two features justify the large difference of SEC values between the industries of both countries (660 kWh/ton_{RM} against 465 kWh/ton_{RM}).

Ham industries

The ham production industries (9 units) are plants of large dimensions with a great number of cold room (on average 17 cold rooms) as presented in Table 3. The type of productive process (several stages) and the long period necessary to obtain the ham requires that these industries have a high number of cold rooms. The chambers where the post-salting, drying and stuffing is carried out differ from the cold rooms because they allow the cooling and heating. It is important to mention that the temperatures needed for the Portuguese ham stuffing stage require auxiliary heating systems, usually electric resistances integrated in the compact unit.

The energy consumption in the Portuguese ham industry is predominantly electric, corresponding on average to 92.5 % of the overall energy consumption. For these industries, the consumption of electric energy is comprised between 336.1 and 2565.6 MWh. The overall compressor power is between 93 and 346 kW.

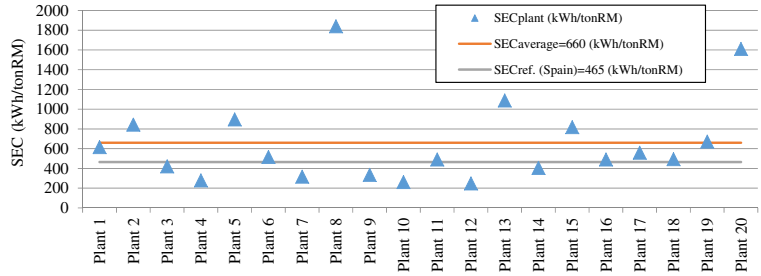


Fig. 7 Values of the specific electricity consumption (SEC) on each Portuguese sausage manufacturing plant visited, the average value and a reference value from Spanish industries (EREN 2008)

Figure 8 shows the individual values of SEC obtained by Eq. (1). The SEC values from these industries are comprised between 310 and 2202 kWh/ton$_{RM}$, with an average value of 1208 kWh/ton$_{RM}$. Despite developing the same productive process and using similar mechanical refrigeration equipment, there is a great difference in the energy consumption between these industries. This fact suggests that some industries are consuming more energy than the one necessary to carry out its activities. In particular, it was observed that plant 1, plant 5 and plant 8 have oversized facilities compared to the raw material processed. As a result, they present the higher SEC values (and lower EVU values). Based on the SEC distribution for this category, the potential reduction of electric energy consumption is estimated around 8 % (708 MWh).

In Fig. 8, a SEC average value of the ham production industries in Spain is also presented, namely 336 kWh/ton$_{RM}$ (EREN 2008). This value is considerably below the average value found for the Portuguese ham industry, 1208 kWh/ton$_{RM}$.

Besides the fact that Spanish industries use a significant percentage of other energy form that can reach around 50 % (EREN 2008). the major difference of specific consumption average values between the two countries is related to the different methodologies in obtaining the environmental conditions used in some production process stages, specifically for ham drying and stuffing.

In Spain, theses stages are carried out in physical environments (cellars) with strong use of natural climacteric conditions and also making use of some thermal energy, contrary to Portugal, where all the production process stages take place in artificial environments. This is the reason why one of the industries visited (plant 3) present a SEC value considerably below its counterparts since, as verified, the drying and stuffing processes are carried out almost without using refrigeration equipment (compact type). They take advantage of the physical environment conditions (cellars) for the maturation process stage and, when necessary, use thermal energy to produce hot water that flows in additional heat exchangers located inside the rooms. If this procedure was applied to the remaining industries of this category, the potential reduction of electricity would be 50 % (4688 MWh).

Table 3 Results for ham manufacturing plants, based on the information collected locally in each plant and the indicators calculated by Eqs. (1) to (3)

Plants	Year of construction	Raw material (ton$_{RM}$)	N.^{er} cold chambers	V_{total} (m^3)	P_{total} (kW)	E (MWh)	SEC (kWh/ton$_{HSCW}$)	EVU (ton$_{HSCW}$/m^3)	EPCV (kW/m^3)
Plant 1	1998	268.3	14	2852	105	406.0	1513	0.07	0.037
Plant 2	1989	1968.6	18	6958	243	1618.8	822	0.17	0.035
Plant 3	1990	1753.7	17	4555	241	543.1	310	0.32	0.059
Plant 4	1996	2116.0	26	10,394	346	2565.6	1212	0.13	0.033
Plant 5	1990	152.6	12	3016	93	336.1	2202	0.05	0.031
Plant 6	1980	585.9	14	2676	109	515.8	880	0.15	0.041
Plant 7	1977	1139.0	23	5174	263	1409.0	1237	0.20	0.051
Plant 8	1995	889.3	21	6938	279	1506.0	1694	0.09	0.040
Plant 9	1990	404.5	15	1699	101	405.1	1001	0.18	0.059

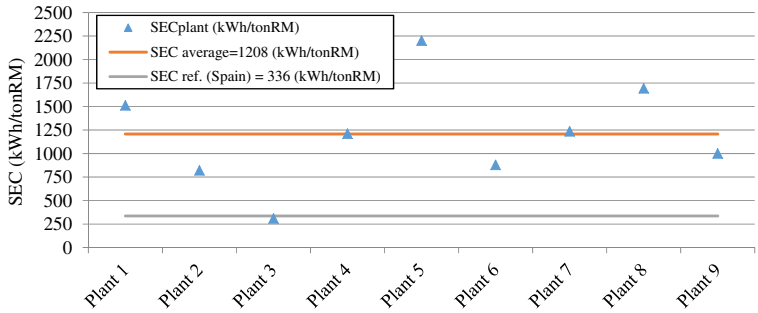


Fig. 8 Values of the specific electricity consumption (SEC) on each Portuguese ham manufacturing plant visited, the average value and a reference value from Spanish industries (EREN 2008)

Comparative study

Next, the general results for the set of the three industry categories will be analysed, including the EVU and EPCV indicators. Regarding the results presented in Tables, 1, 2 and 3, it is possible to observe that sausage houses are the meat industries with the smallest dimension and production. These industries produce on average less quantity of raw material (approximately 255 ton_{RM}/year); they present lower values of compressor power and total volume of cold rooms when compared to the remaining industry categories. Yet, this is the category with the highest representation in the Portuguese meat sector.

The effective volume usage of the cold rooms (EVU), obtained in Eq. (2), varies with the activity and shows large amplitude in the slaughterhouse and sausage house, as observed in Fig. 9. This EVU amplitude is originated either by an incorrect sizing of the number of cold rooms regarding the production or due to a poor management of the production that originates an inadequate use of the cold rooms (in various situations, it was observed that inside the same industry there was more than one cold room working with partial load). However, the consequences bring on a decrease in the industries' energy efficiency and in particular the refrigeration systems. The average value of the EVU is 2.57 ton_{HSCW}/m^3 for slaughterhouses, 0.712 ton_{RM}/m^3 for sausage houses and finally 0.15 ton_{RM}/m^3 for ham production industries. The highest EVU value presented by the slaughterhouses is due to a high transfer of raw material (short period inside the cold room, on average 12 to 24 h), which is a characteristic of the production process in these industries (daily slaughter of animals).

On the other hand, the low occupation rate of the cold room volume in ham production industries, is also related to the type of productive process, in this case, it forces the maintenance of the products inside the rooms during longer periods (around 6 months); therefore, a reduced transfer of raw material is observed inside the cold rooms. Overall, hams are the products with a higher need of cold maintenance and its production is, consequently, the most energy-intensive. Any change in the production process that optimises the permanence in the cold rooms will have a significant impact on the energy cost of this product. In addition, the overall compressor power of the plant per unit of volume (EPCV) depends on the type of activity. Figure 10 shows this indicator for each of the three meat product categories. The average EPCV indicator values for the slaughterhouse and ham

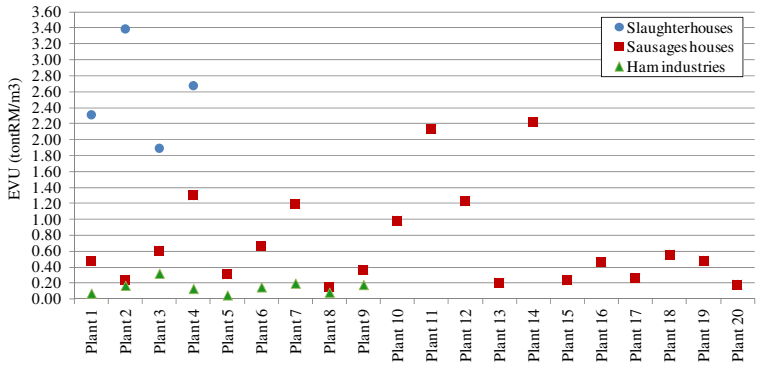


Fig. 9 Utilisation volume rate (EVU) of the cold rooms in the meat industry

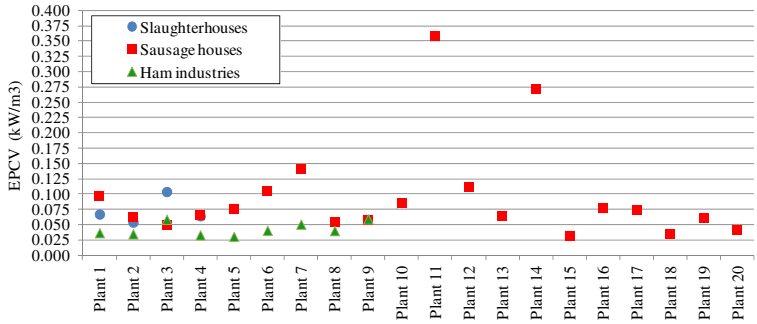


Fig. 10 Electric power absorption rate by the compressors per unit volume (EPCV) of each industry

production industries are 0.072 and 0.043 kW/m^3. The major shifts inside each category correspond to plant 3, for slaughterhouse 8 (discussed above), and to plants 11 and 14 for the sausage houses (since they are the smallest industries—with the lowest cold room volume and lowest quantity of processed raw material). The average value of this indicator for the sausage houses (not taking into consideration the two industries considered unusual) is 0.072 kW/m^3. It is important to notice that this indicator fluctuation shows there is no rule in the sizing process of the mechanical refrigeration systems, which leads to the over-sizing or under-sizing of the systems with implications either in energy consumption or in the maintenance of the inner environmental conditions recommended for the cold rooms.

The proximity of the average indicator value (EPCV) in slaughterhouse and sausage house categories was not expected since cooling requirements are substantially different: slaughterhouses require a higher cooling capacity per unit of cold room volume in order to produce a fast cooling of the animal carcass (from the initial temperature of 38 to 40 °C, to values lower than 7 °C); the sausage houses only deal with cold conservation or drying processes requiring lower cooling capacity.

The fact that a lower average value for the EPCV is registered in ham production industries is not strange: a higher cold room volume of this category (in order to have a high production capacity installed), and a reduced load transfer during the maturation and drying phase result in a lower cooling capacity per unit of volume.

Figure 11 shows the average values of the electric energy annual consumption and SEC values for each category. The average values found for the electric energy consumption were 491.1 MWh (slaughterhouses), 127.1 MWh (sausage houses) and 1034 MWh (ham industries), meaning an average value of 418.6 MWh for the meat sector according with our sample composition.

We can conclude that industries developing meat processing activities, such as sausage houses (660 kWh/ton$_{RM}$) and ham production industries (1208 kWh/ton$_{RM}$) present a SEC value higher than industries that are only devoted to meat production, such

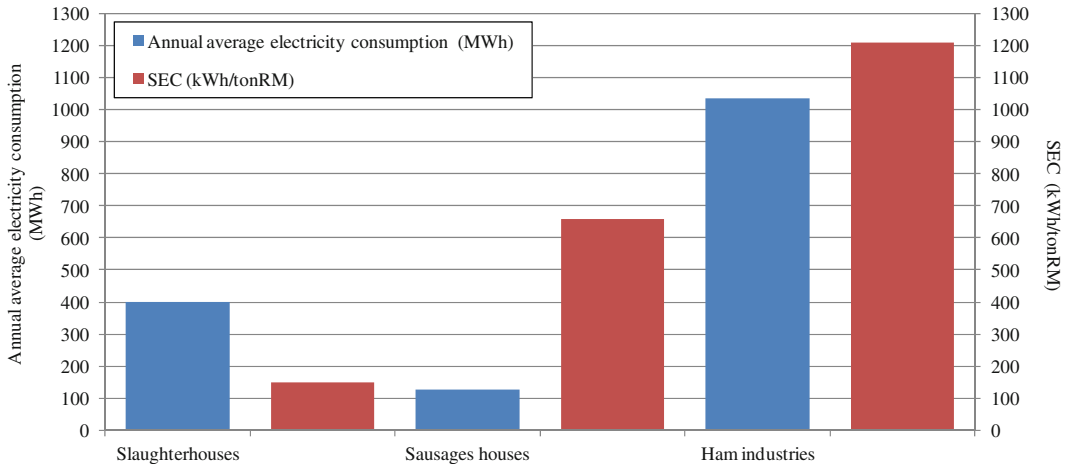


Fig. 11 Annual average of the specific electricity consumption (SEC) of the meat industry

Table 4 Specific electric energy consumption values for the meat sector in other countries

Country	Meat facilities	Product	Energy type	SEC (kWh/ton)	References
UK	Slaughterhouse	Pig carcass	Electrical	36–154	IPPC (2005)
Spain	Slaughterhouse	Pig carcass	Electrical	55–193	MAPA (2005)
OECD	Slaughterhouse	Pig carcass	Electrical	139	UNIDO (2010)
Italy	Slaughterhouse	Pig carcass	Electrical	80–246	EMAS (2015)
Nordic countries	Sausages	Sausages	Electrical	750	IPPC (2006)
Spain	Sausages	Sausages	Electrical	465	EREN (2008)
Nordic countries	Sausages	Ham	Electrical	750–1300	IPPC (2006)
Italy	Sausages	Ham	Electrical	1557	CL (2003)
Spain	Sausages	Ham	Electrical	336	EREN (2008)

as slaughterhouses (149 kWh/ton$_{RM}$). The ham production industries are, therefore, the category that presents simultaneously the highest electric energy consumption and the highest SEC value, being the category that requires greater concern in terms of energy efficiency measures in order to improve the competitiveness of these industries.

Table 4 shows the specific electrical energy consumption (SEC) for several countries found in scientific and technical literature. Globally, the Portuguese meat industry shows SEC values within the range for this kind of industry. The lowest values are found in the Spanish meat industry (sausage and ham manufacturing) resulting from the specific production process using natural conditions and thermal energy on drying process as described above.

Based on the study carried out and related to energy savings, some of the most recommended actions are the following: to improve production process management in order to avoid the use of cold rooms with partial load; to improve the thermal resistance of the building envelope, cold rooms and refrigeration pipes; to avoid condensation group installation in warm places (exposed to solar radiation, in non-ventilated attics or exposed to exhaust air from similar groups). If this recommended energy-saving measures were adopted, an average electrical energy saving of 217 MWh will be expected in the meat sector based on the potential reduction of the estimated electricity consumption (24 % for the slaughterhouses, 13 % for the sausage houses and 8 % for the hams industries) and, accordingly, with the average electric energy consumption in each category. Globally, these energy saving measures applied to every one of the 619 companies in the Portuguese meat sector will contribute for a electrical energy consumption reduction of 8.4 % in the food sector, 0.7 % of the industrial sector and 0.3 % in the national electrical energy consumption.

Conclusions

The energy consumption profile of meat industries is analysed, based on a study of 33 industries in the Central Region of Portugal, divided in three categories: slaughterhouses, sausage houses and ham production industries. The energy balance made to these industries shows that electric energy is the largest form of energy used on average and accounts for 66.5 % of the overall energy consumption in slaughterhouses, 85 % in sausage houses and 92.5 % in ham industries. The average value of the specific electricity consumption (SEC) found for the slaughterhouses was 149 kWh/ton$_{HSCW}$, 660 kWh/ton$_{RM}$ for sausage houses and 1208 kWh/ton$_{RM}$ for ham production industries. These SEC values are higher than the reference values found in scientific and technical literature. Throughout the field study, various aspects were identified which, if improved, will allow a significant improvement in the energy performance of these industries, specially, inadequate infrastructures (building envelope and cold rooms with lower thermal resistance) incorrect management of cold rooms (several operating at partial load), erroneous decision regarding the positioning of some refrigeration components (condensers and compressors sometimes located in non-ventilated attics, exposed to direct solar radiation or in close proximity), refrigeration pipes with damaged insulation among others. On behalf of these industries,

there was an estimation of a potential reduction in the electric energy consumption achieved by carrying out a number of best practices. This potential reduction was estimated in 24 % for the slaughterhouses (corresponding to an energy reduction of 474 MWh), 13 % for the sausage houses (322 MWh) and 8 % for the hams (708 MWh).

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